

MASTER OF SCIENCE IN PRECISION SCIENTIFIC INSTRUMENTATION

Assignment Cover Page

Course: PH6780: Supervised Research

Assessment title: Research Proposal: AI-Assisted Agent-Based Modelling of Collective Foraging Behaviour in Ant Colonies

Student's official name: Chen Yipeng

Student Matriculation No.: G2502593E

Supervisor/Tutor's name: Dr Yong Ee Hou

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Student's signature: Chen Yipeng

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1. Introduction and Background

Collective behaviour in biological systems can emerge without centralised control through repeated local interactions among individuals and their environment. Ant foraging is a well-studied example: individuals possess limited information, yet colonies discover food, recruit workers, establish routes, and allocate workers among tasks. Explaining how simple behavioural mechanisms produce these coordinated outcomes is an important problem in biological physics and non-equilibrium complex systems (Fernández-López et al., 2025; Zhang & Yong, 2023).

Agent-based modelling is suitable for this problem because it preserves the link between individual rules and population-level outcomes. Simulated ants can each have a position, behavioural state, local observations, memory, and transition rules; their interactions can then generate trail formation, food retrieval, and task allocation. Existing models combine correlated random walks, pheromone-mediated communication, memory, and behavioural transitions (Zhang & Yong, 2023), and examine pheromone dynamics in changing or multiple-resource environments (Dodoková et al., 2024; Hartman et al., 2024).

Model construction nevertheless requires choices among movement rules, pheromone responses, task-transition probabilities, and environmental conditions. Their combinations can make exhaustive testing impractical, while manually selected experiments may cover a researcher-dependent subset. Random search and sequential optimisation offer reproducible baselines (Akiba et al., 2019; Bergstra & Bengio, 2012), but do not themselves formulate or revise mechanistic explanations in biological language.

LLM-based agents may assist this exploration by combining hypothesis generation with structured actions and feedback (Yao et al., 2023). Recent preprints have applied this approach to iterative computational research workflows (Lu et al., 2024; Schmidgall et al., 2025), but its value for mechanistic biological modelling remains underexplored. Unrestricted outputs may also be irrelevant or inconsistent unless actions, variables, and evaluation criteria are constrained.

This project therefore proposes a two-layer framework: an interpretable agent-based model of collective foraging and an LLM-based assistant operating at the experiment-design level. The LLM will not control ants or generate simulation code. It will select predefined biologically admissible mechanisms, propose parameterised experiments, receive quantitative summaries, and revise its hypotheses. The biological question is which combinations of pheromone persistence, movement heterogeneity, and task switching support efficient foraging and recovery after food-source relocation. The methodological question is whether LLM-assisted exploration can identify robust configurations more efficiently than conventional search under the same rule space and simulation budget.

2. Review of Related Literature

2.1 Ant Collective Foraging and Agent-Based Modelling

Ant colonies are decentralised biological systems in which collective organisation emerges from local interactions. Agent-based models are useful for studying this process because they retain the connection between individual movement, communication, behavioural roles, and colony-level outcomes.

Zhang and Yong (2023) represented foraging ants as foragers, transporters, and followers. Discrete correlated random walks, pheromone trails, memory, and role transitions support recruitment and task allocation, producing a transition from disordered searching to organised foraging. These explicit mechanisms and measurable outcomes provide the biological foundation for this project.

Other models show that collective foraging depends on environmental conditions and communication; Dodoková et al. (2024) combined off-lattice movement with a spatial field for pheromone diffusion and decay, comparing trail formation and responses to environmental change under alternative navigation mechanisms. This supports examining how finite pheromone persistence affects exploration and adaptation.

Hartman et al. (2024) coupled a stochastic lattice model to a macroscopic description of ant and pheromone dynamics and extended trail analysis to multiple food sources. Their work shows that simplified rules can retain important collective features while remaining mathematically analysable.

Behavioural heterogeneity may help balance exploration and exploitation. Fernández-López et al. (2025) combined observations with a model distinguishing scout-like and recruit-like movement. Although their focal species does not rely primarily on pheromone trails, the study provides independent motivation for testing bounded movement heterogeneity rather than assuming identical ants.

2.2 Limitations of Conventional Model Exploration

Interpretable agent-based models require explicit behavioural rules, but movement, pheromone, memory, and task-transition parameters can interact. Varying one factor at a time may therefore miss mechanisms responsible for a collective outcome.

Manual exploration is useful during development but difficult to reproduce as a comparator. Grid search becomes inefficient as dimensions increase; random search can cover such spaces more effectively (Bergstra & Bengio, 2012), while sequential optimisation uses previous evaluations to select new configurations (Akiba et al., 2019).

These methods are necessary baselines, but primarily optimise numerical objectives rather than state and revise biological explanations. Any advantage of LLM-assisted exploration must therefore be demonstrated using the same rule space, simulation budget, and evaluation conditions.

2.3 LLM-Based Scientific Agents and Iterative Experimentation

The ReAct framework connects language models to external tools through iterative reasoning, action, observation, and revision (Yao et al., 2023). Although not developed for biological modelling, this structure separates language-based proposals from the software that executes and verifies them.

Recent preprints extend this idea to computational research. The AI Scientist generates ideas, modifies code, runs experiments, and prepares reports in selected machine-learning domains (Lu et al., 2024); Agent Laboratory organises agents around literature review, experimentation, and reporting with human feedback (Schmidgall et al., 2025). The cited studies do not evaluate constrained exploration of a mechanistic biological model against conventional optimisation.

A direct transfer would be inappropriate because unrestricted code generation would obscure whether a result arose from a biological hypothesis, an implementation change, or an error. Here, the LLM will be restricted to predefined mechanisms and parameter ranges, while simulation, metric calculation, and validity checks remain conventional software components.

The resulting research gap is a specific, testable question: can an LLM agent use simulation feedback to explore biologically constrained rules more efficiently than conventional search? Ant foraging provides a controlled setting in which mechanisms, actions, budgets, and outcomes can be defined without changing the biological meaning of the underlying model.

3. Specific Aims

The overall aim is to investigate how interpretable individual-level mechanisms influence collective ant foraging and whether an LLM agent can assist their systematic exploration. The biological model remains the primary object of study; the LLM operates only at the experiment-design level.

The project has three specific aims.

1. Develop and validate an interpretable agent-based model of collective ant foraging.

A simulation based on Zhang and Yong (2023) will represent ant positions, behavioural states, local observations, and transition rules. Static tests will assess food discovery, recruitment, trail formation, transport, and task redistribution before food-source relocation is introduced to measure adaptation.

2. Construct a constrained LLM-assisted scientific exploration loop.

A structured interface will allow an LLM agent to propose and revise experiments within a limited space of pheromone, movement, and task-transition mechanisms. Each proposal must include a hypothesis, valid parameters, experimental condition, and expected outcome. The agent will receive only structured summaries and cannot modify executable code.

3. Evaluate LLM-assisted exploration against conventional search strategies.

The LLM-assisted loop will be compared with random search and the Optuna tree-structured Parzen estimator (TPE) (Akiba et al., 2019; Bergstra & Bengio, 2012). All methods will share the same rule space, candidate budget, stochastic replicates, and conditions. Outcomes will include simulator evaluations required to reach a biological target, target discovery within budget, and robustness under held-out conditions.

These aims address two research questions:

1. Which combinations of pheromone persistence, movement heterogeneity, and task-switching rules support efficient collective foraging and rapid recovery after food-source relocation?
2. Under a fixed simulation budget and predefined biologically admissible rule space, can an LLM-based scientific agent identify robust model configurations more efficiently than conventional search strategies?

The corresponding hypotheses are:

H1a: An intermediate pheromone decay rate will reduce median recovery time relative to lower and higher decay rates while retaining at least 80% of the validated baseline's static food retrieval efficiency.

H1b: With mean movement persistence held constant, a colony with bounded scout-like and recruit-like movement heterogeneity will have a shorter median recovery time than a homogeneous colony.

H2: Within an equal budget of 30 candidate opportunities, LLM-assisted exploration will require fewer simulator evaluations than random search to identify a configuration that meets the predefined food-retrieval and post-relocation recovery target.

H1a and H1b will be evaluated separately so that pheromone persistence and movement heterogeneity are not confounded; task-switching parameters will remain part of the exploratory rule space without a directional hypothesis. H2 will be rejected if the LLM-assisted method does not demonstrate a consistent advantage over random search. Comparison with TPE will be a secondary test without a prespecified superiority claim and will indicate whether any benefit is specific to language-based mechanistic hypothesis revision rather than adaptive search in general.

4. Significance of the Proposed Project

The project is primarily a study of collective biological behaviour. By extending an established three-role foraging model (Zhang & Yong, 2023) to environmental change, it will examine how local movement, communication, and task transitions affect recovery when a previously useful trail becomes outdated.

The biological contribution will compare interpretable mechanisms that balance exploration and exploitation. Persistent pheromone can stabilise a route but delay adaptation; movement heterogeneity can preserve exploratory workers; and task-transition rules redistribute activity as conditions change. Their effects will be assessed within one simulation environment using food retrieval, trail dynamics, and recovery time.

Methodologically, the project will test whether an LLM agent can help explore mechanistic simulations by stating a hypothesis, selecting a permitted experiment, interpreting structured results, and revising its next proposal. A validated interface will prevent code modification and keep each parameter change traceable.

A controlled comparison with random search and sequential optimisation is needed because adaptive selection is not unique to LLM agents. Identical search spaces and budgets will test whether language-based hypothesis revision provides a measurable advantage. Failure to improve efficiency, invalid proposals, or poor held-out performance would instead identify practical limitations.

The intended contribution is not autonomous discovery of biological laws, but an evaluated workflow for exploring a bounded set of interpretable hypotheses. Conclusions will remain limited to the implemented model, rule space, and environmental conditions.

5. Intended Approach

5.1 Agent-Based Foraging Model

The project will implement a two-dimensional agent-based simulation based primarily on the behavioural framework of Zhang and Yong (2023). Each ant will be represented by its position, movement direction, behavioural state, local observations, and limited memory. The principal behavioural states will represent searching for food, following recruitment information, and transporting food to the nest. State transitions will be stochastic and dependent on locally available information rather than centralised instructions.

The environment will contain a nest, a food source, and a spatial pheromone field. Searching ants will perform correlated random walks until they encounter food or detect a sufficiently strong pheromone signal. Ants returning with food will deposit pheromone, which will diffuse and decay over time. These local processes should generate collective outcomes including food discovery, recruitment, trail formation, transport, and redistribution among behavioural states.

The pheromone concentration $P(\mathbf{x}, t)$ will be represented by a diffusion–decay field with local deposition sources (Dodoková et al., 2024):

$$\frac{\partial P(\mathbf{x}, t)}{\partial t} = D_P \nabla^2 P(\mathbf{x}, t) - \lambda P(\mathbf{x}, t) + S(\mathbf{x}, t), \quad (1)$$

where D_P is the pheromone diffusion coefficient, λ is the decay rate, and $S(\mathbf{x}, t)$ represents pheromone deposited by returning ants. Equation (1) will be discretised in space and time using an explicit finite-difference update, with deposition added to the lattice cells occupied by depositing ants. The initial field will satisfy $P(\mathbf{x}, 0) = 0$, with $P = 0$ at the outer boundary of the simulated domain. The grid spacing and time step will be chosen to satisfy the stability condition of the explicit diffusion–decay update.

The baseline model will first be tested in a static single-food-source condition. The implementation will be considered adequate for subsequent experiments only if it consistently produces food discovery, recruitment, transport to the nest, and a transition from dispersed searching to organised foraging. Basic software tests will also check population conservation, valid state transitions, parameter bounds, and reproducibility under recorded random seeds.

After baseline validation, the food source will be relocated during the simulation. This dynamic condition will test whether the colony continues to follow an outdated trail or redistributes activity towards renewed exploration. Multiple-food-source environments will be treated as an optional extension rather than a required component of the project.

5.2 Biologically Admissible Rule Space

The exploration process will be limited to three classes of interpretable mechanisms:

1. **Pheromone persistence:** deposition, diffusion, decay, and sensory-threshold parameters that determine how long information about a food source remains influential;
2. **Movement heterogeneity:** bounded differences in movement persistence or exploratory tendency among ants, motivated by the distinction between scout-like and recruit-like behaviour (Fernández-López et al., 2025);
3. **Task switching:** probabilities or thresholds controlling transitions among searching, following, and transporting states.

All permitted mechanisms and parameter ranges will be defined before the comparative exploration experiments begin. The LLM agent and conventional search methods will use exactly the same space. This prevents the LLM from gaining an advantage by introducing mechanisms unavailable to the baseline methods.

5.3 LLM-Assisted Exploration Loop

The LLM agent will interact with the simulator through a structured tool interface. At each iteration, it will receive the current objective, the permitted mechanisms and parameter ranges, and summaries of previous experiments. It must then return:

- a testable behavioural hypothesis;

- the selected mechanism or mechanism combination;
- parameter values within the permitted ranges;
- the expected effect on collective behaviour; and
- a brief justification for the proposed experiment.

The validator will reject missing fields, out-of-range values, and unpermitted mechanisms. The LLM cannot access or modify simulator code; after each valid run, it receives only summary statistics and retains, revises, or rejects its hypothesis. Prompts, model versions, tool definitions, proposals, and results will be recorded. Each requested proposal counts as one candidate opportunity. One repair prompt may correct schema or range errors without changing the hypothesis; a proposal that remains invalid is recorded as failed and is not simulated.

5.4 Controlled Comparison

The LLM-assisted method will be compared with two conventional strategies:

1. random sampling from the same rule and parameter space; and
2. sequential optimisation using Optuna's TPE sampler and the same numerical evaluation criteria.

Table 1: Comparison of the three exploration strategies.

Method	Use of feedback and hypotheses	Selection rule
Random search	No previous-result feedback and no explicit hypothesis	Random sampling
TPE	Numerical outcomes from previous configurations; no mechanistic hypothesis	Density-based sequential optimisation
LLM agent	Structured simulation summaries and explicit hypothesis revision	Language-guided experiment selection

The planned comparison will use ten independent campaigns per method and 30 candidate opportunities per campaign. Each valid configuration will be evaluated using the same fixed set of 20 stochastic seeds. Random-search and TPE sampler seeds will vary between campaigns, and the LLM agent will begin each campaign with an empty experimental history. TPE will minimise median recovery time, with a penalty applied to configurations that fail the static-efficiency constraint. A timing-only pilot may reduce

the campaign or replicate count before comparative experiments begin; any reduction will be fixed in advance and applied equally to all methods.

The primary biological outcome for each stochastic replicate will be recovery time following food-source relocation. Let t_r denote the relocation time, T_{post} the post-relocation duration, R_{pre} the mean retrieval rate during the final 20% of the pre-relocation phase, and $R(t)$ the instantaneous retrieval rate. With a trailing averaging window $w = 0.05T_{\text{post}}$, the smoothed retrieval rate and elapsed recovery time are defined as

$$\begin{aligned} w &= 0.05T_{\text{post}}, \\ I &= [t_r + w, t_r + T_{\text{post}} - w], \\ \tilde{R}(t) &= \frac{1}{w} \int_{t-w}^t R(u) \, du, \quad t \geq t_r + w, \\ \tau_{\text{rec}} &= \inf \left\{ t - t_r : t \in I, \min_{s \in [t, t+w]} \tilde{R}(s) \geq 0.8R_{\text{pre}} \right\}. \end{aligned} \tag{2}$$

The integral will be implemented as the corresponding discrete moving average. If the set in Equation (2) is empty, the run will be assigned $\tau_{\text{rec}} = T_{\text{post}}$ and also recorded as a non-recovery. A qualifying configuration must reduce median recovery time by at least 20% relative to the validated homogeneous baseline while retaining at least 80% of its static food retrieval efficiency.

Secondary biological measures will include food discovery time, trail formation time, variability in food retrieval, pheromone-trail persistence, and the proportion of ants in each behavioural state. Method-level outcomes will include the number of evaluated configurations required to reach the biological target, the proportion of campaigns that find a qualifying configuration within the budget, the invalid-proposal rate, and performance under held-out environmental conditions.

5.5 Statistical Analysis and Robustness

Simulation outcomes will be summarised across replicates using medians, uncertainty intervals, and effect sizes. Exploration methods will be compared at campaign level by target-discovery rate and evaluations required to reach the target. Configurations selected by each method will then be tested with held-out food locations, colony sizes, and random seeds to assess robustness.

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